

DATASET A – CORRECTION NOTICE

MIMIC-III → Diabetes 130-US Hospital Readmissions

This notice replaces the Dataset A description in the Final Project brief. All other datasets unchanged.

1. What Happened and Why Dataset A Has Changed

The original Dataset, the MIMIC-III clinical database (~46,000 admissions), has a Kaggle mirror, which is the PhysioNet demo subset and contains only 129 hospital admissions – a sample size that is analytically unusable for the regression, logistic, and cross-validation tasks in the project.

Accessing the full MIMIC-III dataset requires completing CITI human subjects training and submitting a credentialing application to PhysioNet, a process that takes 2–6 weeks. This is not feasible within the project timeline.

Action required

Groups originally assigned Dataset A must switch to the replacement dataset described below. No analytical work done on the 129-row demo is wasted – the variable structure is similar and EDA skills transfer directly. Groups that have already begun should re-run their data audit on the new dataset; the time cost is small.

2. Replacement Dataset A – Diabetes 130-US Hospital Readmissions (1999–2008)

Attribute	Detail
Full name	Diabetes 130-US Hospitals for Years 1999–2008
Primary source	UCI Machine Learning Repository https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008
Kaggle mirror	Search “diabetes 130-us hospitals readmission” on kaggle.com Direct URL: kaggle.com/datasets/brandao/diabetes
Rows / Columns	101,766 hospital encounters × 50 variables (47 features + 3 identifiers)
Access	Free, no registration required from UCI. Kaggle requires free account.
File format	diabetic_data.csv + IDs_mapping.csv (lookup table for coded fields)
Published in	Strack et al. (2014). Impact of HbA1c Measurement on Hospital Readmission Rates. BioMed Research International.
License	CC BY 4.0 (UCI) – free for academic use with citation

3. Revised Primary Analytical Question

The primary question for Dataset A is revised to match the new dataset:

What patient characteristics, clinical complexity indicators, and diabetes management factors best predict early hospital readmission (within 30 days), and does the effect of HbA1c measurement on readmission risk differ across admission types?

This question maps directly onto the project tasks:

Project Part	How the primary question maps on
Part 1 – EDA	Audit the 50-variable dataset; diagnose MNAR missingness in weight and medical_specialty; classify ordinal vs. nominal variables (age, max_glu_serum, A1Cresult)
Part 2 – Inference	Compare readmission rates across admission types (chi-square); compare time_in_hospital distributions by readmission status (Welch's t or Mann-Whitney)
Part 3 – MLR	Outcome: time_in_hospital (continuous, discrete). Predictors: num_medications, number_diagnoses, A1Cresult, age midpoint, number_inpatient. Interaction: A1Cresult × admission_type
Part 4 – Logistic	Binary outcome: readmitted_30 (1 = readmitted <30 days, ~11% of rows). Predictors: same as Part 3 plus diag_1 category
Part 5 – Regularization	Include all 24 medication columns as dummies → high-dimensional predictor space where Lasso naturally performs variable selection
Part 6 – Pitfalls	Simpson's paradox: aggregate readmission rate by race, then stratify by admission type. Patient-level clustering (multiple encounters per patient) violates independence – discuss as a validity threat

4. Difficulty Factors (unchanged from original brief)

The replacement dataset preserves all the analytical challenges described in the original brief. In several respects it is more analytically demanding.

Challenge	Detail
Class imbalance	Only ~11% of encounters result in early readmission (<30 days) – identical to the MIMIC-III mortality rate. Default threshold of 0.5 will be suboptimal; students must optimise threshold (Part 4.3b).
MNAR missingness	Weight is missing for 97% of rows. Missingness is not random – lighter and heavier patients may be under-recorded differently. This is a strong MNAR example for Part 1.3.
High-dimensional categoricals	24 medication columns × 4 levels each. One-hot encoding produces a wide matrix ideal for Lasso regularization in Part 5.
Repeated measures	One patient_nbr can appear across multiple encounters. Standard regression assumes independence. Students must acknowledge this as a validity threat in Part 6.2.
ICD-9 code grouping	diag_1/2/3 contain raw ICD-9 codes (~900 unique values). Must be grouped into clinical categories before use as predictors. See Appendix A.

Ordinal vs. continuous	age is stored as 10-year bins ([30-40) etc.). Students must justify whether to treat it as ordinal, midpoint-numeric, or dummy-encoded. Each choice has different implications.
Multiple outcome framings	readmitted has three levels. Students must justify their binarisation choice (30-day cutoff vs. any readmission) and acknowledge the information lost.

5. Dataset-specific guidance

Part 1.1

Diabetes 130-US: Load with `na_values="?"` — missing values are encoded as literal `?`, not blank cells. The three key missing columns are `weight` (~97%, MNAR: recording correlates with clinical severity), `medical_specialty` (~49%, MAR: correlates with admission type — test before asserting), and `payer_code` (~40%). Do not classify `AlCresult = "None"` or `max_glu_serum = "None"` as missing — these mean the test was not ordered, which is a substantive clinical state.

Part 2.1

Diabetes 130-US: Compute the CI for mean `time_in_hospital` for the full sample and separately by early readmission status. The variable is discrete and bounded (1–14); invoke the CLT explicitly to justify the t-CI. Also compute a bootstrap CI for the median, which is more informative than the mean for this bounded, right-skewed distribution.

Part 3

Diabetes 130-US: Outcome = `time_in_hospital` (apply square-root transformation if residual plots show heteroskedasticity — justify with diagnostics). Key predictors: `number_diagnoses`, `num_medications`, `number_inpatient`, age midpoint, `AlCresult` (ordinal-encoded), admission type (grouped). Interaction to test: `AlCresult × admission_type_grouped`.

Part 4

Diabetes 130-US: Define `readmitted_30 = 1` if `readmitted == "<30"`, 0 otherwise (~11% positive class). Before modelling, remove rows where `discharge_disposition_id` indicates in-hospital death (value 11) or hospice (values 13, 14, 19, 20, 21) — these patients cannot be readmitted and their inclusion corrupts the outcome.

Part 6.1

Diabetes 130-US: Aggregate early readmission rate by `race`; then stratify by `admission_type_grouped`. African American patients are over-represented in emergency admissions, which carry higher readmission rates regardless of race. Controlling for admission type will narrow or eliminate the crude racial gap — discuss whether this means race is irrelevant or whether admission type is itself a downstream consequence of unequal access.

ICD-9 Code Grouping for diag_1 / diag_2 / diag_3

Raw ICD-9 codes in diag_1 have ~900 unique values and cannot be used directly as predictors. Group them into the 9 clinical categories used in the original Strack et al. (2014) paper before including them in any model. Apply the same grouping to diag_2 and diag_3.

Category label	ICD-9 code range / rule	Clinical meaning
Circulatory	390–459, 785	Heart disease, hypertension, stroke
Respiratory	460–519, 786	Pneumonia, COPD, asthma
Digestive	520–579, 787	GI disorders
Diabetes	250.xx	Diabetes mellitus (all subtypes)
Injury	800–999	Trauma and poisoning
Musculoskeletal	710–739	Arthritis, fractures
Genitourinary	580–629, 788	Renal and urinary disorders
Neoplasms	140–239	Cancer
Other	All remaining numeric codes + codes starting with E or V	External causes, supplementary codes

Suggested Python implementation:

```
def map_icd9(code):
    """Map a raw ICD-9 code string to one of 9 clinical categories."""
    if pd.isna(code) or code in ('?', ''):
        return 'Unknown'
    code = str(code).strip()
    if code.startswith('V') or code.startswith('E'):
        return 'Other'
    try:
        num = float(code)
    except ValueError:
        return 'Other'
    if 390 <= num <= 459 or num == 785: return 'Circulatory'
    if 460 <= num <= 519 or num == 786: return 'Respiratory'
    if 520 <= num <= 579 or num == 787: return 'Digestive'
    if 250 <= num < 251: return 'Diabetes'
    if 800 <= num <= 999: return 'Injury'
    if 710 <= num <= 739: return 'Musculoskeletal'
    if 580 <= num <= 629 or num == 788: return 'Genitourinary'
    if 140 <= num <= 239: return 'Neoplasms'
    return 'Other'

dff['diag_1_cat'] = dff['diag_1'].apply(map_icd9)
dff['diag_2_cat'] = dff['diag_2'].apply(map_icd9)
dff['diag_3_cat'] = dff['diag_3'].apply(map_icd9)
```

6. Original vs. Replacement Dataset at a Glance

	MIMIC-III (original, unusable)	Diabetes 130-US (replacement)
Rows available without credential	129 – demo only	101,766 – full dataset
Access	PhysioNet credentialing (2–6 weeks)	Free, instant (UCI / Kaggle)
Binary outcome	In-hospital mortality (~11% positive)	Readmission <30 days (~11% positive)
Key continuous outcome	Length of stay (days)	time_in_hospital (days)
Severity score	SAPS-II	number_diagnoses + A1Cresult
Medication variables	Several	24 medication columns (ideal for Lasso)
ICD codes	ICD-9 via DIAGNOSES_ICD.csv (complex join)	diag_1/2/3 in same row (simple)
Class imbalance	~11% mortality	~11% early readmission
Difficulty factors	MNAR vitals, clustering, skewed LOS	MNAR weight, repeated encounters, ordinal age, high-dim medications